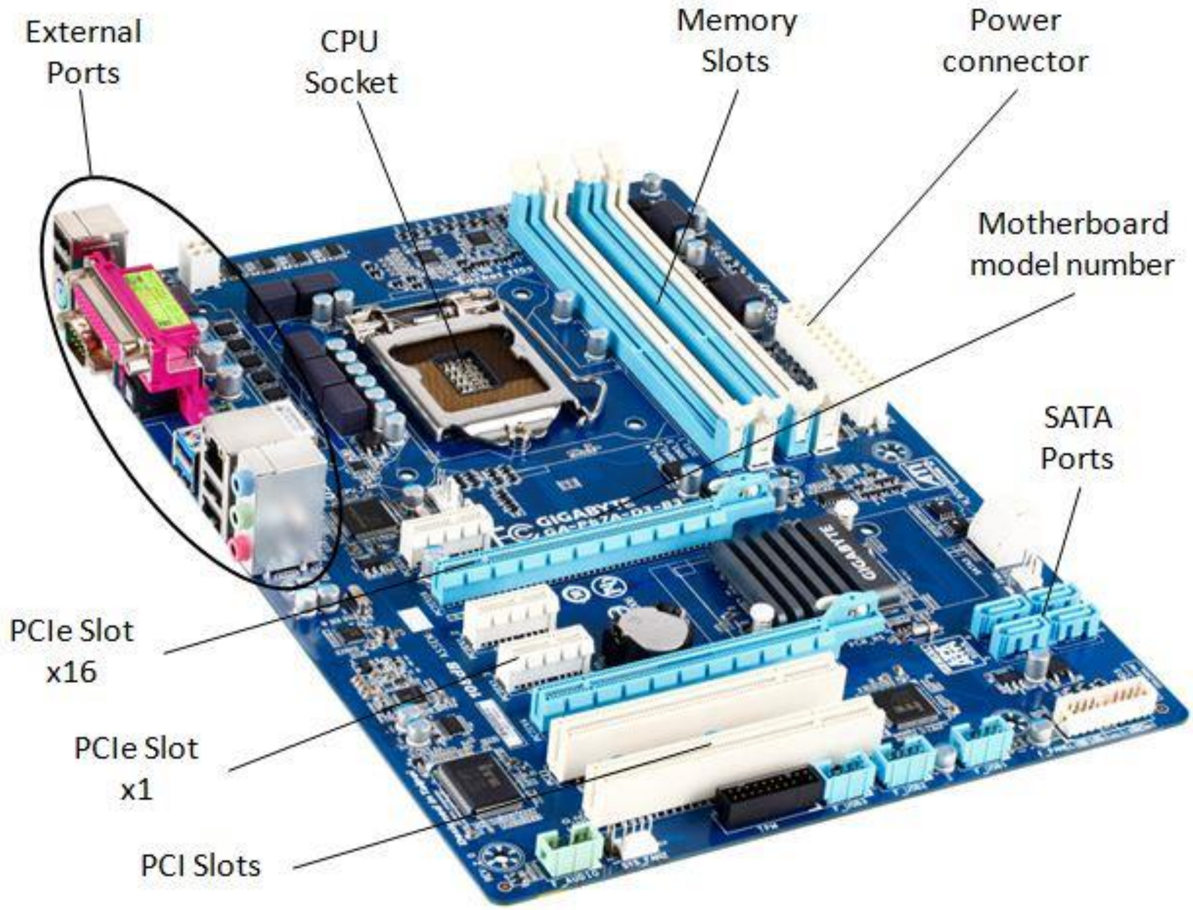
DAY-2

2.1 Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



DAY-2

2.2 Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

CPU Socket Type: LGA1151

Manufacturer: Intel

Compatible CPU Model: Intel Core i7-9700K

DAY-2

2.3 Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit):

The CPU is the brain of the computer. It processes instructions, performs calculations, and controls the overall operation of programs. When you use an application, the CPU executes the commands needed to make it work.

Chipset:

The chipset is a group of controllers on the motherboard that manages communication between the CPU and other hardware components, such as the RAM, SSD/HDD, USB devices, graphics card, and network adapter. It ensures that data is transferred efficiently between these components.

1. You click the Spotify icon.
2. The CPU receives the command and starts executing the instructions to launch the application.
3. The chipset helps the CPU communicate with the SSD or HDD to load Spotify's files into RAM.
4. The CPU processes the program while the chipset coordinates data transfer between the CPU, RAM, storage, and network hardware.
5. When you play a song, the chipset helps the network adapter receive music data from the internet, and the CPU processes it so the audio can be sent to your speakers or headphones.

DAY-2

2.4 Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.